

ASYMPTOTIC RELATIVE EFFICIENCIES OF TESTS BASED ON THE METHOD OF n RANKINGS WHEN THE UNDERLYING DISTRIBUTIONS ARE DISCRETE

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SUMMARY

The problem is considered of testing the null hypothesis of equal treatment effects. The underlying distributions are assumed to be discrete. The asymptotic efficiencies of the midrank Friedman test, the average Van der Waerden scores test, the midrank Van der Waerden scores test, and two tests based on randomized ranks, are compared with the asymptotic efficiency of the test based on the variance ratio statistic.

1. Introduction

Van Elteren and Noether (1959) were the first authors to compare the asymptotic efficiencies of the Friedman and F-tests. Subsequent comparisons have been made by Sen (1967, 1968 and 1972). Lemmer, Stoker and Reinach (1968) assumed that there are $b \geq 1$ observations per cell in the randomized block design. They used the method of n rankings to derive a class of test statistics which can be used as a basis of a test of the null hypothesis of equal treatment effects. Among other generalizations, this class yields generalizations of the Friedman statistic and of the statistic based on Van der Waerden scores. Lemmer, Stoker and Reinach (1968) studied the asymptotic relative efficiencies (a.r.e.'s) of tests based on these statistics relative to the test based on the variance ratio statistic.

The above authors all assumed the underlying distributions to be continuous. Markham (1980) considered the situation in which the underlying distributions are discrete. It was shown that a test of the null hypothesis may be based either on a member of a class of average scores test statistics, or on a member of a class of midrank scores test statistics. The asymptotic efficiencies of certain members of these classes will be compared with the asymptotic efficiency of the F-test. The asymptotic efficiencies of the randomized Friedman test and the randomized Van der Waerden scores test will also be con-

sidered.

A listing of the test statistics under consideration is given in the appendix.

2. Assumptions and a.r.e. values

Consider the randomized block design having k treatments, n blocks, and one observation per cell. For $i=1, \dots, n$, $j=1, \dots, k$, let X_{ij} be the random variable associated with the outcome in cell (i, j) . Let $d(r; t_{ij})$ denote the probability, $P(X_{ij}=r)$, that the random variable X_{ij} assumes the value $r \in C$, where C is the set of integers $\{\dots, -1, 0, 1, \dots\}$. The quantity t_{ij} is the parameter under consideration.

The null hypothesis to be tested is, $H_0: t_{i1} = \dots = t_{ik} = t_i$ for each i , while the sequence of local alternatives considered is, for each i , $K_n: t_{ij} = t_i + n^{-1/2} \delta_j$, δ_j finite, constant, $j=1, \dots, k$, $\sum_j \delta_j = 0$.

The following theorem shows that for large n a test of H_0 may be based on the variance ratio statistic.

Theorem

(a) If H_0 holds, then as $n \rightarrow \infty$, $F_n \xrightarrow{D} \chi^2(k-1)$, where

$F_n = n \sum_j (x_{.j} - x_{..})^2 / [\sum_i \sum_j (x_{ij} - x_{i.} - x_{.j} + x_{..})^2 / (n-1)(k-1)]$, $x_{.j} = n^{-1} \sum_i x_{ij}$, with analogous definitions holding for $x_{i.}$ and $x_{..}$.

(b) Suppose that $\mu_i = \sum_r r d(r; t_i)$, and $\lim_n \sigma_n^{-2} = n^{-1} \sum_i \sum_r (r - \mu_i)^2 d(r; t_i)$ exists and is finite.

Then, if K_n holds, as $n \rightarrow \infty$, $F_n \xrightarrow{D} \chi'^2(k-1, \lambda'_F)$, where

$$\lambda'_F = (\sum_j \delta_j^2) \lambda_F, \text{ with } \lambda_F = (\lim_n \sigma_n^{-2})^{-1} \lim_n [n^{-1} \sum_i \sum_r r d'(r; t_i)]^2, \text{ and } d'(r; t_i)$$

is the partial derivative of $d(r; t_i)$ with respect to t_i .

The result of part (b) holds provided that for each $i=1, \dots, n$, $j=1, \dots, k$, and $r \in C$ the partial derivative $d'(r; t_{ij})$ is continuous in t_{ij} on some open neighbourhood, J_i , of the finite parameter t_i . In addition, for each i and j , as $R \rightarrow \infty$, $\{\sum_{r \in R} r d'(r; t_{ij})\}$ must converge uniformly on J_i to a finite function which is continuous in t_{ij} on J_i .

Proof

The proof is analogous to that given on p. 274 of Puri and Sen (1971) for continuous distributions. ||

In principle the theorem allows a comparison to be made of the asymptotic performances of the F-test and tests based on members of the classes of randomized scores statistics, average scores statistics and midranks scores statistics that were considered by Markham (1980). Unfortunately, as can be seen from the expression for λ_F , an exact

numerical comparison of the tests is impossible. Consequently, as in Markham (1980), attention will be restricted to the approximation obtained by assuming that for each i , $t_{ij} = t_j$.

The discrete distributions considered, viz. the Poisson, binomial, negative binomial and discrete uniform distributions, are the same as those considered by Conover and Kemp (1976). The forms of the p.m.f.'s used are given below.

- (i) Poisson distribution : $d(r; t) = e^{-t} t^r / r!$, $r = 0, 1, 2, \dots$.
- (ii) Binomial distribution : $d(r; \pi) = \binom{N}{r} \pi^r (1-\pi)^{N-r}$, $r = 0, 1, \dots, N$, since, given N , t is equivalent to π .
- (iii) Negative binomial distribution : $d(r; \pi) = \binom{m+r-1}{r} \pi^m (1-\pi)^r$, $0 < \pi < 1$, $m > 0$, $r = 0, 1, 2, \dots$, since, given m , t is equivalent to π .
- (iv) Discrete uniform distribution, (Conover and Kemp (1976)):
 $d(r; t) = N^{-1} + \{r - (N+1)/2\} \epsilon$, $r = 1, \dots, N$, $|\epsilon| < 2/(N(N-1))$, where
 $t = (n+1)/2 + \epsilon N(N^2 - 1)/12$. For fixed N , $t \rightarrow (N+1)/2$ as $\epsilon \rightarrow 0$.

To confirm that the regularity conditions stated in the theorem, as well as those given in Markham (1980), are satisfied by the four distributions under consideration, use may be made of results given in Conover and Kemp (1976); of a result on p. 223 of Apostol (1974); of the Weierstrass M-test; and of routine mathematical analysis.

The asymptotic efficiencies of the randomized Friedman test and the midrank Friedman test relative to that of the F-test are shown in Tables 1 to 4.

TABLE 1
Asymptotic relative efficiencies of Friedman-type tests when sampling
from Poisson distributions

Randomized ranks				Midranks			
k	$t=0,4$	1,0	3,0	k	$t=0,4$	1,0	3,0
2	0,413	0,549	0,609	2	0,868	0,793	0,731
3	0,465	0,617	0,686	3	0,882	0,832	0,791
4	0,496	0,658	0,731	4	0,890	0,852	0,824
5	0,516	0,686	0,762	5	0,894	0,865	0,846
6	0,531	0,705	0,784	6	0,897	0,874	0,861
∞	0,620	0,823	0,914				

TABLE 2
Asymptotic relative efficiencies of Friedman-type tests when sampling from binomial distributions

Randomized ranks							Midranks						
	N=8			N=16				N=8			N=16		
k	$\pi=0,05$	0,15	0,25	0,05	0,07	0,25	k	$\pi=0,05$	0,15	0,25	0,05	0,07	0,25
2	0,423	0,573	0,603	0,530	0,561	0,620	2	0,882	0,807	0,783	0,820	0,800	0,740
3	0,476	0,645	0,679	0,596	0,631	0,697	3	0,896	0,849	0,835	0,852	0,840	0,801
4	0,507	0,688	0,724	0,636	0,673	0,744	4	0,903	0,872	0,863	0,869	0,862	0,836
5	0,528	0,717	0,754	0,663	0,701	0,775	5	0,907	0,887	0,881	0,880	0,875	0,858
6	0,544	0,737	0,775	0,682	0,721	0,797	6	0,910	0,896	0,894	0,887	0,885	0,873
∞	0,634	0,860	0,905	0,795	0,842	0,930							

TABLE 3*

Asymptotic relative efficiencies of Friedman-type tests when sampling from negative binomial distributions

Randomized ranks							Midranks						
	m=2			m=4				m=2			m=4		
k	$\pi=0,95$	0,73	0,50	0,95	0,83	0,50	k	$\pi=0,95$	0,73	0,50	0,95	0,83	0,50
2	0,165	0,463	0,538	0,278	0,503	0,587	2	0,932	0,753	0,660	0,899	0,771	0,661
3	0,186	0,521	0,605	0,313	0,565	0,661	3	0,935	0,781	0,711	0,905	0,804	0,724
4	0,198	0,556	0,645	0,334	0,603	0,705	4	0,936	0,797	0,739	0,908	0,821	0,760
5	0,207	0,579	0,672	0,348	0,628	0,734	5	0,936	0,806	0,758	0,910	0,832	0,784
6	0,213	0,595	0,691	0,358	0,646	0,755	6	0,937	0,812		0,911	0,839	
∞	0,248	0,694	0,807	0,417	0,754	0,881							

*Blank entries correspond to values for which the convergence was too slow.

TABLE 4
Asymptotic relative efficiencies of Friedman-type tests when sampling
from discrete uniform distributions

Randomized ranks				Midranks			
k	N=8	12	16	k	N=8	12	16
2	0,656	0,662	0,664	2	0,750	0,722	0,708
3	0,738	0,745	0,747	3	0,818	0,796	0,785
4	0,788	0,794	0,797	4	0,857	0,839	0,829
5	0,820	0,828	0,830	5	0,882	0,867	0,859
6	0,844	0,851	0,854	6	0,900	0,886	0,879
∞	0,984	0,993	0,996				

Thus far we have considered the situation in which the underlying distributions are truly discrete. In practice, however, the underlying distributions are usually continuous, and ties are caused by a lack of precision in the measuring instruments. In such an event we shall refer to the underlying distributions as discrete continuous distributions.

An attempt will be made to answer the following questions for the above situation.

How does the skewness of the discrete continuous distribution affect the a.r.e. values?

Are the a.r.e. values the same for all symmetric discrete continuous distributions?

How does lack of precision of the measuring instruments affect the asymptotic performances of the respective tests?

It is not known what effect the variance has on the a.r.e. Therefore the scale parameters will be so chosen as to ensure that the variance of each underlying continuous distribution has the value unity. In each case t will be chosen to be the location parameter of the underlying continuous distribution, and attention will be restricted to the null hypothesis $H_0: t=0$.

The discrete normal distribution $d(r; t) = \Phi(r + \frac{1}{2} - t) - \Phi(r - \frac{1}{2} - t)$, $r \in C$, will be chosen as the reference distribution. The other distributions considered are the discrete exponential distribution derived from the exponential distribution with location parameter t , and two members of the class of discrete exponential power distributions defined by $d(r; t) = F(r + \frac{1}{2}; t) - F(r - \frac{1}{2}; t)$, $r \in C$, $|t| < \frac{1}{2}$, where

$$F(u; t) = F'(u; t) = w(\kappa) \exp[-c(\kappa)|u - t|^{2/(1+\kappa)}], \quad -\infty < u < \infty, -1 < \kappa < 1,$$

and the expressions for $w(\kappa)$ and $c(\kappa)$ are given on p. 157 of Box and Tiao (1973).

The distributions chosen were the leptokurtic, $\kappa=1$, and the platykurtic, $\kappa=-\frac{1}{2}$, distributions. The first of these is the discrete double exponential distribution.

The variances of the above discrete continuous distributions are 1,08; 1,16; 1,07 and 1,02 respectively.

Reasoning analogous to that employed for the discrete distributions shows that the required regularity conditions are satisfied by the above discrete continuous distributions. The calculated a.r.e. values can be found in Table 5.

TABLE 5*

Asymptotic relative efficiencies of Friedman-type tests when sampling from discrete continuous distributions

Randomized ranks					Midranks				
k	Normal $\sigma=1$	Expon. $\sigma=1$	Expon. Power		k	Normal $\sigma=1$	Expon. $\sigma=1$	Expon. Power	
			$\kappa=1,0$	$-0,5$				$\kappa=1,0$	$-0,5$
2	0,587	0,924	0,696	0,571	2	0,805	1,368	1,040	0,659
3	0,660	1,039	0,783	0,642	3	0,851	1,430	1,094	0,718
4	0,704	1,108	0,835	0,685	4	0,877	1,464	1,122	0,751
5	0,734	1,154	0,870	0,714	5	0,892	1,485	1,140	0,773
6	0,755	1,187	0,895	0,734	6	0,903	1,499		
∞	0,880	1,385	1,044	0,856					

*Blank entries correspond to values for which the convergence was too slow.

TABLE 6

Asymptotic relative efficiencies of the Friedman test when sampling from continuous distributions

k	Normal $\sigma=1$	Expon. $\sigma=1$	Expon. Power	
			$\kappa=1,0$	$-0,5$
2	0,637	2,000	1,000	0,582
3	0,716	2,250	1,125	0,655
4	0,764	2,400	1,200	0,698
5	0,796	2,500	1,250	0,727
6	0,819	2,571	1,286	0,748
∞	0,955	3,000	1,500	0,873

In order to determine how lack of precision of the measuring instruments affects the asymptotic performances of the tests, suppose that perfect measuring instruments were available. A test of the null hypothesis of equal treatment effects could then be based on the Friedman statistic. Table 6 contains the a.r.e. values of a test of the null hypothesis based on the Friedman statistic relative to the F-test.

3. Discussion of results

The asymptotic efficiencies of the tests based on randomized Van der Waerden scores, average Van der Waerden scores, and midrank Van der Waerden scores, were also computed. It was found that there is only a marginal difference between the asymptotic efficiencies of tests based on Friedman scores and those based on Van der Waerden scores.

In all cases there is a considerable loss of efficiency if the randomized scores test is used instead of the average scores or midrank scores test.

Discrete distributions

In general the midrank Van der Waerden scores test seems to be the best of the (asymptotically) distribution-free tests considered. It should, however, be noted that for the discrete distributions considered, the F-test is asymptotically superior to all of the above-mentioned tests.

Discrete continuous distributions

- (i) It is not possible to state that either the midrank Friedman test, the average Van der Waerden scores test, or the midrank Van der Waerden scores test performs consistently better than the other two tests.
- (ii) The a.r.e.'s calculated for the skew discrete exponential distribution are higher than those calculated for the discrete normal distribution.
- (iii) The a.r.e. values associated with the leptokurtic discrete exponential power distribution with $\kappa = -\frac{1}{2}$, and the leptokurtic discrete uniform distribution, are lower than those associated with the discrete normal distribution. The opposite is true for the platykurtic discrete double exponential distribution. The question remains as to whether this result holds for all symmetric discrete continuous distributions.
- (iv) From Tables 5 and 6 it can be seen that the a.r.e. values corresponding to the midrank Friedman test are closer to unity than are the a.r.e. values associated with the Friedman test. It would thus appear as though a loss of precision results in a.r.e. values that are closer to unity than are the a.r.e. values obtained for perfect measuring instruments.

Appendix

Consider the randomized block design having k treatments, n blocks, and one observation per cell.

The variance ratio statistic is

$$F_n = n \sum_{j=1}^k (x_{.j} - x_{..})^2 / [\sum_{i=1}^n \sum_{j=1}^k (x_{ij} - x_{i.} - x_{.j} + x_{..})^2 / (n-1)(k-1)],$$

where x_{ij} is the observation in cell (i, j) , and $x_{.j}$ denotes $n^{-1} \sum_i x_{ij}$. Similar definitions

apply for the other terms.

The rank associated with x_{ij} is

$$r_{ij} = \sum_{j'=1}^k u(x_{ij} - x_{ij'}), \quad j=1, \dots, k, i=1, \dots, n,$$

where $u(x) = 1$ if $x \geq 0$, and $u(x) = 0$ if $x < 0$. Let $\tau_{i1}, \dots, \tau_{ig(i)}$ denote the lengths of the groups of equal observations (ties) in the i -th block, and, for $\nu=1, \dots, g_i (=g(i))$, define

$$T_{i0}=0, T_{i\nu} = \sum_{\mu=1}^{\nu} \tau_{i\mu}.$$

The midrank Friedman statistic is

$$\bar{Q}_n = [12(k-1) / (n\sigma^2)] \sum_j [\sum_i (\bar{r}_{ij} - (k+1)/2)]^2,$$

where

$$\sigma^2 = k(k^2-1) - n^{-1} \sum_i \sum_{\nu=1}^{g(i)} \tau_{i\nu} (\tau_{i\nu}^2 - 1),$$

and

$$\bar{r}_{ij} = \tau_{i\nu}^{-1} \sum_{\mu=T(i, \nu-1)+1}^{T(i\nu)} \mu \quad \text{if } T_{i, \nu-1} < r_{ij} \leq T_{i\nu}$$

is the midrank associated with x_{ij} .

The average Van der Waerden scores statistic is

$$\bar{Q}_{Vn} = (k-1)(\sum_i \sum_j a_{ij}^2)^{-1} \sum_j [\sum_i \bar{a}_{ij}]^2,$$

where

$$\bar{a}_{ij} = \tau_{i\nu}^{-1} \sum_{\mu=T(i, \nu-1)+1}^{T(i, \nu)} \Psi(\mu/(k+1)) \quad \text{if } T_{i, \nu-1} < r_{ij} \leq T_{i\nu}.$$

The midrank Van der Waerden scores statistic is

$$Q_{mVn} = (k-1)(nk\sigma_{mn}^2)^{-1} \sum_j \left[\sum_i (\Psi(\bar{r}_{ij}/(k+1)) - \bar{a}_{mi}) \right]^2,$$

where

$$\bar{a}_{mi} = k^{-1} \sum_j \Psi(\bar{r}_{ij}/(k+1)),$$

and

$$\sigma_{mn}^2 = (nk)^{-1} \sum_i \sum_j \left[\Psi(\bar{r}_{ij}/(k+1)) - \bar{a}_{mi} \right]^2.$$

The randomized rank associated with x_{ij} is given by

$$r_{ij}^* = \sum_j u(x_{ij} + u_{ij} - x_{ij'} - u_{ij'}) \quad j=1, \dots, k, \quad i=1, \dots, n,$$

where u_{ij} , $i=1, \dots, n$, $j=1, \dots, k$ are independent drawings from the uniform distribution on the interval (0,1).

The randomized Friedman statistic is

$$Q_n^* = (12/nk(k+1)) \sum_j \left[\sum_i (r_{ij}^* - (k+1)/2) \right]^2,$$

while the randomized Van der Waerden scores statistic is

$$Q_{Vn}^* = (k-1)(nk\sigma_V^2)^{-1} \sum_j \left[\sum_i \Psi(r_{ij}^*/(k+1)) \right]^2,$$

where

$$\sigma_V^2 = k^{-1} \sum_j \Psi^2(j/(k+1)).$$

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